Introduction & Setup

PSY 410: Data Science for Psychology

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Why are we here?

Psychology has a data problem

In 2015, a team of 270 researchers tried to replicate 100 published psychology studies.

. . .

Only 36% produced the same results.

. . .

This wasn't fraud. These were real labs, following published methods, using real data.

So what went wrong?

Many things went wrong

- Analyses that couldn't be reproduced (even by the original authors)
- Data that was cleaned by hand with no record of what changed
- Figures that obscured rather than revealed patterns
- Code that only worked on one person's laptop

. . .

The replication crisis isn't just a statistics problem. It's a **workflow** problem.

This course is about the workflow

Every technical skill we learn this quarter serves the same goal:

. . .

> Start with raw data. End with a clear, honest, reproducible story.

. . .

That means learning to:

- **Import** data without breaking it
- **Transform** it transparently (with code, not clicks)
- **Visualize** it to find patterns — and catch errors
- **Communicate** what you found so others can verify it

About this course

What we'll learn:

- Data visualization with ggplot2
- Data transformation with dplyr
- Data tidying with tidyr
- Reproducible reports with Quarto

Why it matters:

- Psychology is increasingly data-driven
- These skills make your work reproducible
- They transfer to any career that touches data

The old way vs. the new way

Point-and-click (SPSS, Excel)

- Hard to reproduce
- Error-prone
- Limited visualizations
- Doesn't scale

Code-based (R)

- Fully reproducible
- Transparent & shareable

- Unlimited customization
- Handles any data size

Why R specifically?

What you need	R delivers
Free	Open source — no licenses, ever
Built for data	Created by statisticians, not software engineers
Psychology packages	`psych`, `lavaan`, `lme4`, `brms`
Publication-quality figures	`ggplot2` — industry-leading
Reproducibility	Quarto integration (we'll learn this)

Getting set up

What you need to install

1. **R** — the programming language
   - Download from cloud.r-project.org
2. **RStudio** — the interface we'll use to write R
   - Download from posit.co/download/rstudio-desktop

. . .

> **!** Important
>
> Install R **first**, then RStudio. RStudio needs R to work!

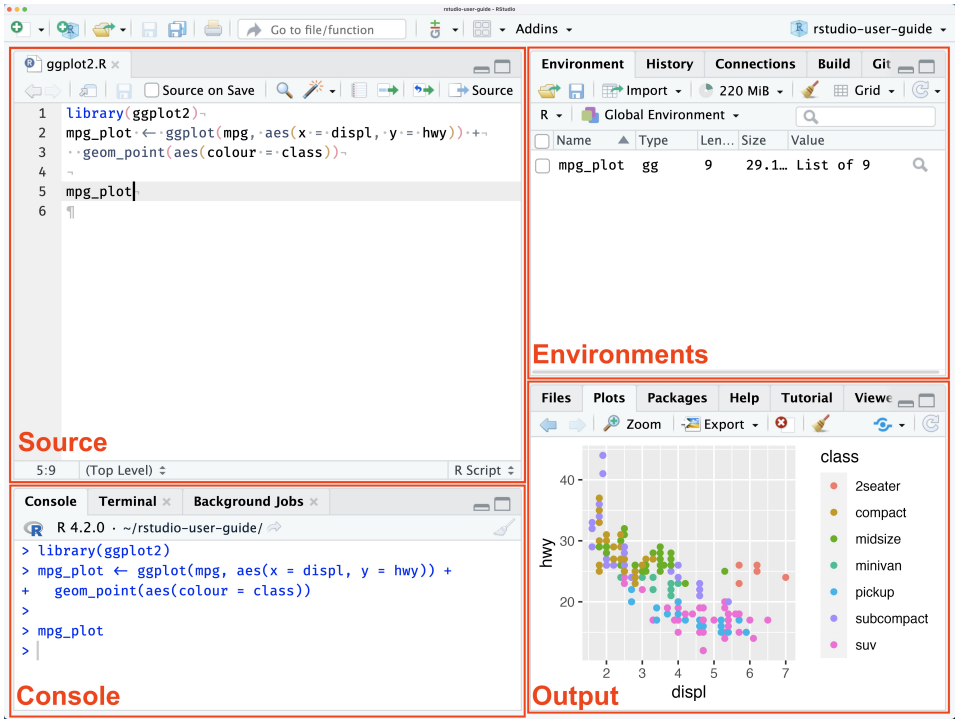
R vs RStudio

Think of it like this:

- **R** is the engine of a car
- **RStudio** is the dashboard, steering wheel, and GPS

You *could* drive with just an engine… but why would you?

The RStudio interface



The four panes: Source (top-left), Console (bottom-left), Environment (top-right), Files/Plots/Help (bottom-right)

The four panes

Pane	What it does
Source (top-left)	Write and edit your code files
Console (bottom-left)	Run code interactively, see output
Environment (top-right)	See your data and objects
Files/Plots/Help (bottom-right)	Navigate files, view plots, get help

Let's try it: The Console

Type in the Console and press Enter:

```r
2 + 2
```

. . .

```r
2 + 2
```

```
[1] 4
```

Congrats — you just ran R code!

The assignment operator

In R, we use `<-` to assign values to objects:

```r
x <- 10
```

Think of it as an arrow pointing left: "put 10 into x"

. . .

```r
# You can use objects in calculations
x * 2
x + x
```

. . .

> Tip
>
> **Keyboard shortcut:** Alt + - (Windows) or Option + - (Mac)

Organizing your work

Two mental models

The filing cabinet

The laundry basket

Two mental models

The filing cabinet

Every piece of paper has a **place** where it *lives*.

You navigate **to** that place to find it.

```
Documents/
  psy410/
      data/
      scripts/
  thesis/
      data/
      drafts/
```

The laundry basket

Everything is in **one pile**.

You **search** to find it.

Google, Spotlight, Finder search, "Recent files"…

Why coding requires the filing cabinet

When you write:

```r
survey <- read.csv("data/raw/survey_responses.csv")
```

You're giving directions: "Start here. Go into `data`. Then into `raw`. Find `survey_responses.csv`."

. . .

If the file isn't exactly there, the code **breaks**. No fuzzy matching. No "did you mean…?"

. . .

> **Important**
>
> This is why we need to learn directory structure — even if it feels foreign.

Why use RStudio Projects?

Without projects:

- Files scattered everywhere
- `setwd()` nightmares
- "It works on my computer"
- Lost work when switching tasks

With projects:

- Everything in one folder
- Paths just work
- Share the whole folder
- Easy to switch between projects

. . .

> Tip
>
> **The golden rule:** Someone else should be able to run your code and get the same results. That "someone else" includes Future You — who has forgotten everything.

Creating a project

1. File → New Project…
2. Choose "New Directory" → "New Project"
3. Name it (e.g., "psy410")
4. Pick a location (Documents folder is fine)
5. Click "Create Project"

. . .

You'll see a `.Rproj` file appear — this is your project file. From now on, double-click it to open your project.

A standard project structure

```
psy410/
  psy410.Rproj
  data/
      raw/           # Original data (READ ONLY)
      clean/         # Processed data
```

```
scripts/
    01_clean.R     # Data cleaning
    02_analyze.R   # Main analysis
    03_visualize.R # Figures
output/
     figures/       # Saved plots
```

A standard project structure

Two principles:

1. **Raw data is sacred** — never modify original files
2. **Outputs are disposable** — you can always regenerate them from code

Naming things

Good names make code **self-documenting**. Bad names create confusion and bugs.

Do this:

- `reaction_time`
- `mean_anxiety`
- `survey_clean.csv`
- `01_clean_data.R`

Not this:

- `x1`, `temp`, `foo`
- `AvgAnx` (hard to read)
- `data_final_v2_REAL.csv`
- `stuff.R`, `untitled3.R`

The test: Can someone unfamiliar with your project understand what a variable contains or what a file does?

R Scripts

R scripts (`.R` files) are where you write and save your code.

To create one:

1. File → New File → R Script
2. Or press Ctrl/Cmd + Shift + N

. . .

Always save your scripts! The console history disappears.

Running code from a script

- **Run one line:** Put cursor on line, press Ctrl/Cmd + Enter
- **Run selection:** Highlight code, press Ctrl/Cmd + Enter
- **Run entire script:** Ctrl/Cmd + Shift + Enter

. . .

> Tip
>
> You'll use Ctrl/Cmd + Enter constantly. Memorize it!

Comments

Use # to write comments — notes for yourself and others:

```
# This calculates the mean age of participants
mean_age <- mean(ages)

# Anything after # is ignored by R
x <- 10  # this assigns 10 to x
```

Comments are essential. Your future self will thank you.

Your first real task

Installing packages

R's power comes from **packages** — bundles of functions others have written.

```
# Install a package (do this once)
install.packages("tidyverse")
```

```
# Load a package (do this every session)
library(tidyverse)
```

The tidyverse

The `tidyverse` is actually a collection of packages:

Package	Purpose
`ggplot2`	Data visualization
`dplyr`	Data manipulation
`tidyr`	Data tidying
`readr`	Reading data files
`tibble`	Modern data frames
`stringr`	String manipulation
`forcats`	Working with factors
`purrr`	Functional programming

We'll use most of these throughout the course.

Let's look at some data

After loading tidyverse, you have access to built-in datasets:

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
# mpg is a dataset about fuel economy
mpg
```

```
# A tibble: 234 x 11
   manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
   <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
```

```
 1 audi         a4           1.8  1999     4 auto~ f        18    29 p     comp~
 2 audi         a4           1.8  1999     4 manu~ f        21    29 p     comp~
 3 audi         a4           2    2008     4 manu~ f        20    31 p     comp~
 4 audi         a4           2    2008     4 auto~ f        21    30 p     comp~
 5 audi         a4           2.8  1999     6 auto~ f        16    26 p     comp~
 6 audi         a4           2.8  1999     6 manu~ f        18    26 p     comp~
 7 audi         a4           3.1  2008     6 auto~ f        18    27 p     comp~
 8 audi         a4 quattro   1.8  1999     4 manu~ 4        18    26 p     comp~
 9 audi         a4 quattro   1.8  1999     4 auto~ 4        16    25 p     comp~
10 audi         a4 quattro   2    2008     4 manu~ 4        20    28 p     comp~
# i 224 more rows
```

Your turn!

1. Create a new RStudio Project called "psy410"
2. Inside it, create three folders: `data`, `scripts`, `output`
3. Create a new R script in `scripts/` and save it as `01_practice.R`
4. Install and load the tidyverse
5. Run `glimpse(mpg)` and `glimpse(diamonds)`
6. Save your script!

Getting help

When you're stuck

1. **Use `?`** — e.g., `?mean` opens the help page
2. **Google it** — seriously, everyone does this
3. **Stack Overflow** — most R questions are answered there
4. **R4DS book** — r4ds.hadley.nz
5. **Ask me** — that's what I'm here for!

Reading help pages

```
?mean
```

Help pages include:

- **Description** — what the function does
- **Usage** — how to call it

- **Arguments** — what inputs it takes
- **Examples** — working code you can run

The examples section is gold — run them!

Error messages

Errors will happen. A lot. That's normal.

```
# This will error - can you spot why?
maen(c(1, 2, 3))
```

```
Error in maen(c(1, 2, 3)): could not find function "maen"
```

. . .

Read the error message carefully — R is trying to help you.

Why no AI? (Yet)

The bicycle vs. the motorcycle

Steve Jobs called the computer a **"bicycle for the mind"** — it amplifies your pedaling, but *you* provide the balance and direction.

Generative AI is more like a **motorcycle**. It provides the engine. But if you don't know how to ride, you crash faster and harder.

Read Cat Hicks: "Cognitive helmets for AI bicycles"

The metacognition trap

AI gives you a working answer immediately. It *feels* like you solved it.

But you didn't build the **mental model** of *why* it works.

When the AI hallucinates (which it will), **you cannot debug it.** You are stranded.

Using AI now = skipping the gym but expecting to get strong.

Building the helmet first

Before we get on the motorcycle, we need a **cognitive helmet**:

1. **Syntax literacy** — reading code as fluently as English
2. **Debugging resilience** — emotional regulation when things break
3. **Strategic friction** — breaking big problems into small pieces

That's what the many short assignments in this course build. The goal: **you** are the pilot, not the passenger.

Wrapping up

Before next class

Read:

- R4DS Ch 2: Workflow basics
- R4DS Ch 6: Workflow: scripts and projects

Do:

- Install R and RStudio
- Create your psy410 project with `data/`, `scripts/`, and `output/` folders
- Install the tidyverse package
- Poke around! Try things! Break stuff!

Key takeaways

1. **Psychology needs better data practices** — that's what this course builds
2. **R is a tool** — like learning any tool, it takes practice
3. **Organization matters** — filing cabinet, not laundry basket
4. **Name things clearly** — your future self will thank you
5. **Errors are learning opportunities** — read them, Google them

The one thing to remember

Every skill you learn in this course is a brick in a wall between you and the replication crisis.

See you Wednesday for your first visualization!